



Constraints on Higgs-charm couplings at CMS

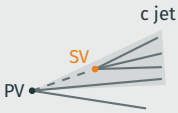
Chirayu Gupta
on behalf of the CMS Collaboration
Vrije Universiteit Brussel

Higgs 2026, Louvain-la-Neuve, 28 October 2026

Charm tagging: telling c jets from b and light jets

Charm signature

c-hadron mass and lifetime lie **between light and b**:
- secondary vertex (SV), like b jets
- but closer to the primary vertex (PV): easily confused with light jets



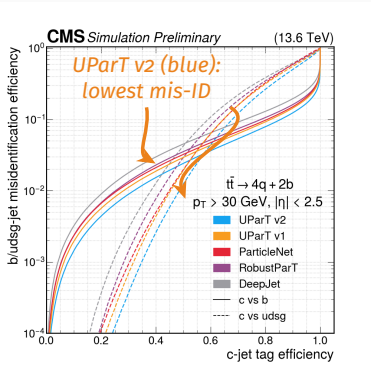
Algorithm evolution at CMS

Likelihood / BDT
JP: track impact parameters
CSV: SV + track variables

Deep NN
DeepCSV: multi-class b/c/light
DeepJet: constituent sequences

Particle cloud
ParticleNet: graph network
UParT: transformer

Run 1 → Run 3: richer architectures, inputs closer to raw particles



$t\bar{t}$ simulation, 13.6 TeV; solid: c vs b, dashed: c vs light

Used in this talk: DeepJet (VH resolved, cH), ParticleNet (VH merged, $t\bar{t}$H), DeepDoubleX (boosted H)

Direct H → $c\bar{c}$: VH(H → $c\bar{c}$)

PRL 131 (2023) 061801, CMS-HIG-21-008

Cleanest signature: W/Z → 0, 1, 2 leptons
V+jets and $t\bar{t}$ normalised in control regions

0L: Z→ $\nu\nu$ **1L: W→ $\ell\nu$** **2L: Z→ $\ell\ell$**

below $p_T^H = 300$ GeV above

Resolved: two jets

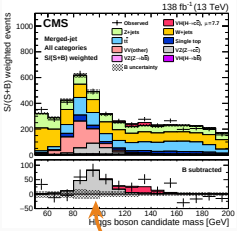
>95% of signal
two DeepJet c-tagged jets
fit to a BDT discriminant

Merged: one large-R jet

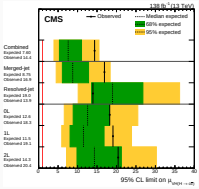
<5% of signal, but **less background**
ParticleNet $c\bar{c}$ tag (3 purities)

fit to $m($

$\mu < 14\,(7.6)$ $1.1 < |\kappa_c| < 5.5\ (|\kappa_c| < 3.4)$
95% CL, observed (expected)



VZ(Z→ $c\bar{c}$) at 5.7σ validates the method



Most sensitive single channel of Run 2;